MOHAMED BIN ZAYED UNIVERSITY OF ARTIFICIAL INTELLIGENCE

Advanced Mathematical Statistics

Lecture 9: Asymptotic Theory of Testing

SDS8102 | SDS8102 | PhD course

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The power of a test near the null

Two tests can both detect every fixed alternative eventually.

How much data does each test need?

- **Local power:** let the alternative approach the null.
- **Relative efficiency:** match the level and target power.
- **Optimality:** find the largest power any test can attain.

In plain language. The root-n scale makes the signal comparable to sampling noise.

Source: van der Vaart (1998), §§14.1–14.3

Notation and local alternatives

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_\theta$, with $\theta \in \Theta \subset \mathbb{R}^d$ and interior reference value θ_0.

- **Empirical average:** $P_n f = n^{-1} \sum_i f(X_i)$.

- **Local parameter and sample law:** keep h fixed:

$$\theta_n(h) = \theta_0 + h/\sqrt{n}, \qquad P_{n,h} = P_{\theta_n(h)}^{\otimes n}.$$

  Every fixed h is admissible for all sufficiently large n.

- **Expectations:** $\mathbb{E}_0$ concerns one observation at θ_0; $\mathbb{E}_{n,h}$ concerns the full sample under $P_{n,h}$.

Φ is the standard normal cdf; $z_q = \Phi^{-1}(q)$. Throughout, $0 < \alpha < 1$.

Convergence in distribution

Weak convergence $Y_n \rightsquigarrow Y$ in $\mathbb{R}^k$ means

$$\mathbb{E}f(Y_n) \to \mathbb{E}f(Y) \quad \text{for every bounded continuous } f : \mathbb{R}^k \to \mathbb{R}.$$

- **Changing laws:** the left expectation is $\mathbb{E}_{n,h} f(Y_n)$ when working under $P_{n,h}$. A common probability space is unnecessary.
- **Normal shorthand:** $Y_n \rightsquigarrow N_k(m, \Sigma)$ specifies the limiting law.
- **Probability notation:** $\xrightarrow{\mathbb{P}}$ denotes convergence in probability; $o_{\mathbb{P}}(1)$ is a vanishing error; $O_{\mathbb{P}}(1)$ means bounded in probability, or tight.

Estimating equations and influence functions

For an estimating function ψ_θ, set $\Psi_n(\theta) = P_n\psi_\theta$ and $\Psi(\theta) = \mathbb{E}_0\psi_\theta(X)$.

- **Linear expansion:** for a consistent root, assume

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = n^{-1/2}\sum_i \varphi(X_i) + o_{\mathbb{P}}(1), \qquad \varphi = -A^{-1}\psi_{\theta_0}.$$

  Here $A = \dot{\Psi}(\theta_0)$ is nonsingular.

- **Covariance:** assume $\mathbb{E}_0\varphi = 0$ and finite second moments:

$$V = \mathbb{E}_0\varphi\varphi^T = A^{-1}BA^{-T}, \qquad B = \mathrm{Var}_0(\psi_{\theta_0}(X)).$$

- **Local alternatives:** To calculate local power, derive the statistic's limiting distribution under $\theta_n = \theta_0 + h/\sqrt{n}$.

Level and asymptotic level

A test $\phi_n : \mathcal{X}^n \to [0,1]$ gives the conditional rejection probability. Its power is $\beta_n(\theta) = \mathbb{E}_\theta \phi_n$.

- **Level α:**

$$\sup_{\theta \in \Theta_0} \beta_n(\theta) \leq \alpha.$$

- **Uniform asymptotic level:**

$$\limsup_n \sup_{\theta \in \Theta_0} \beta_n(\theta) \leq \alpha.$$

- **Pointwise asymptotic level:**

$$\limsup_n \beta_n(\theta) \leq \alpha \quad \text{for each fixed null } \theta.$$

For a simple null the last two coincide. Under a composite null, the worst-case parameter can move with n.

Source: van der Vaart (1998), §14.1

Fixed alternatives and consistency

Let $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, with known σ. Test $\theta \leq 0$ against $\theta > 0$.

- **Rejection rule:** $Z_n = \sqrt{n}\overline{X}_n/\sigma > z_{1-\alpha}$.
- **Exact power:**

$$\beta_n(\theta) = 1 - \Phi\left(z_{1-\alpha} - \sqrt{n}\theta/\sigma\right).$$

  It increases in θ and equals α at zero.

- **Consistency:** $\beta_n(\theta) \to 1$ at every fixed $\theta > 0$.

Quick check. If two tests are consistent, does this imply they need the same sample size?

Three scales of alternatives

In the normal example, let $\theta_n = hn^{-\gamma}$ with $h > 0$.

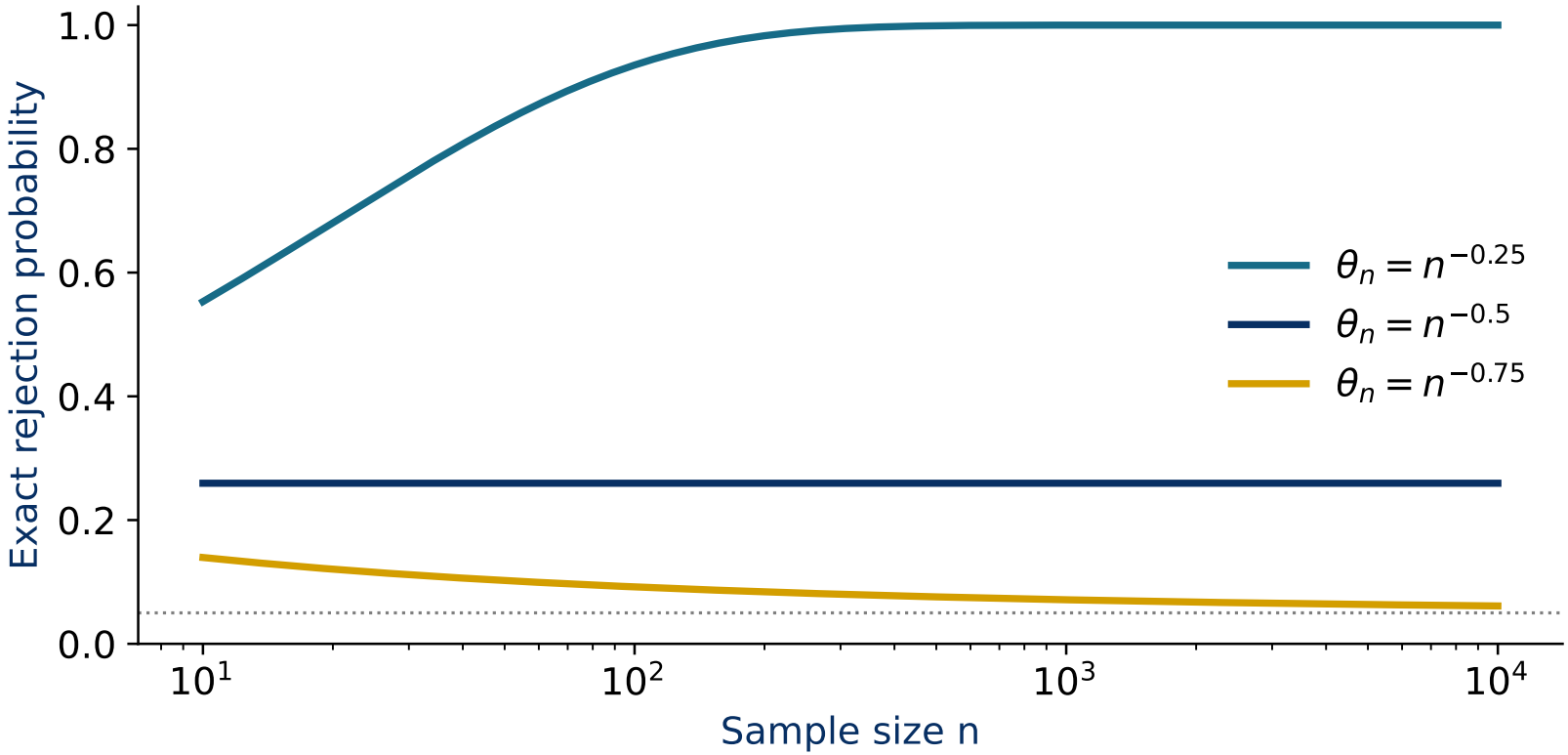
$$Z_n \sim N\left(\frac{h}{\sigma} n^{1/2-\gamma}, 1\right).$$

Rate	Standardized shift	Limiting power
$\gamma < 1/2$	tends to $+\infty$	1
$\gamma = 1/2$	h/σ	$1 - \Phi(z_{1-\alpha} - h/\sigma)$
$\gamma > 1/2$	tends to 0	α

The $n^{-1/2}$ scale matches the standard error in a regular model.

Source: van der Vaart (1998), §14.1; normal illustration

Separation rates in the normal model



$\alpha = .05$, known $\sigma = 1$, and $h = 1$. These curves use the exact normal power function.

Asymptotic local power

Definition 1

When the limit exists, the local asymptotic power is

$$\beta(h) = \lim_n \mathbb{E}_{n,h}\phi_n, \qquad h \text{ fixed.}$$

- **Normal mean:** $\beta(h) = 1 - \Phi(z_{1-\alpha} - h/\sigma)$.
- **At the boundary:** $\beta(0) = \alpha$ for this test.
- **At a local alternative:** $h > 0$ describes $\theta_0 + h/\sqrt{n}$.

Source: van der Vaart (1998), §14.1

Portmanteau lemma

Lemma 2 (Portmanteau)

Suppose random vectors $X_n \rightsquigarrow X$ in $\mathbb{R}^k$. The underlying probability spaces and laws may change with n.

1. ***Closed sets:*** *for closed F,*

$$\limsup_n \mathbb{P}(X_n \in F) \leq \mathbb{P}(X \in F).$$

2. ***Continuity sets:*** *if $\mathbb{P}(X \in \partial D) = 0$,*

$$\mathbb{P}(X_n \in D) \to \mathbb{P}(X \in D).$$

Here D is Borel and ∂D is its boundary.

This standard lemma is stated without proof.

Source: van der Vaart (1998), Lemma 2.2; companion notes, local-power section

A general local power theorem

Theorem 3 (Normal local limit)

For each fixed scalar h, assume under $P_{n,h}$ that

$$T_n \rightsquigarrow N(\kappa h, 1), \qquad c_n \xrightarrow{\mathbb{P}} z_{1-\alpha}, \quad \kappa > 0.$$

Then $\phi_n = \mathbf{1}\{T_n > c_n\}$ satisfies

$$\mathbb{E}_{n,h}\phi_n \to 1 - \Phi(z_{1-\alpha} - \kappa h).$$

- **Slope κ:** standardized signal per unit of local change.
- **Required limit:** the normal approximation must hold under $P_{n,h}$, including moving alternatives.

Source: van der Vaart (1998), §14.1; formulation for this lecture

Proof of the local power theorem

Proof of Theorem 3.

Fix h, let $c = z_{1-\alpha}$, and take $Z \sim N(\kappa h, 1)$.

- **Limit:** Lemma 22, parts 4 and 2, gives under $P_{n,h}$

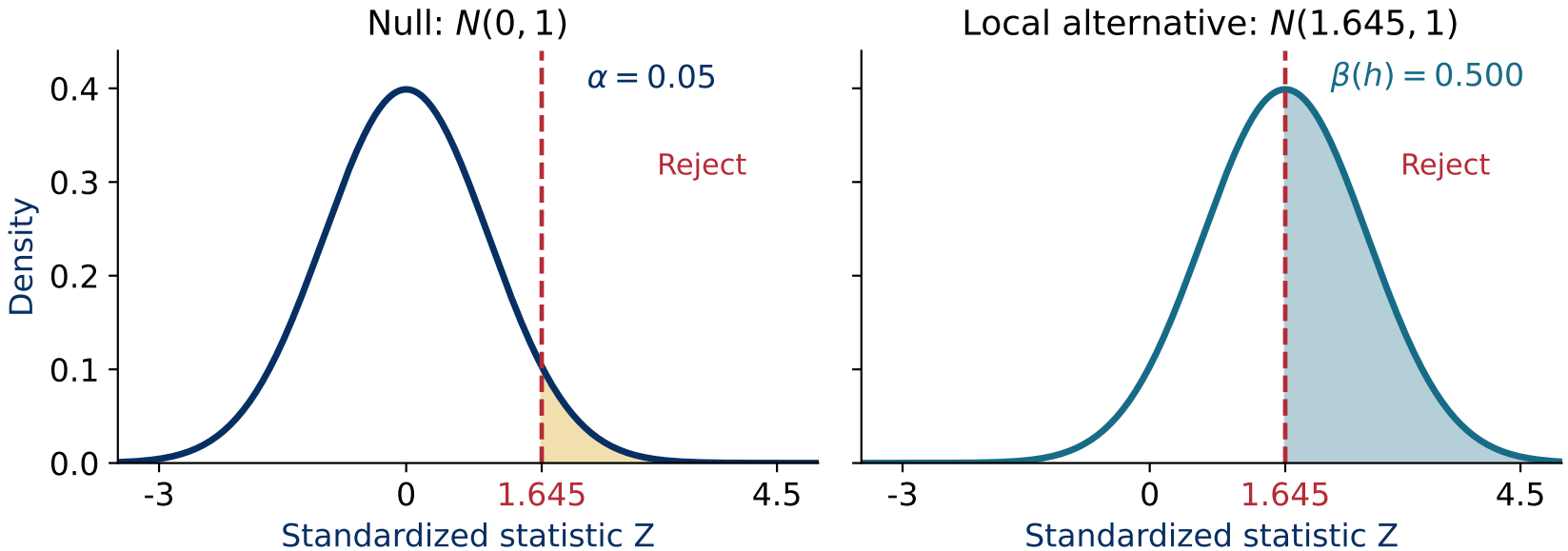
$$X_n = T_n - c_n \rightsquigarrow X = Z - c.$$

- **Boundary:** for $D = (0, \infty)$, $\partial D = \{0\}$ and $\mathbb{P}(X = 0) = 0$.
- **Rejection probability:** Lemma 2, part 2 gives

$$P_{n,h}(T_n > c_n) \to \mathbb{P}(Z > c) = 1 - \Phi(c - \kappa h). \qquad \square$$

Quick check. Which set and which zero-mass boundary justify this probability limit?

A picture of local power



Same rule: reject for $Z > 1.645$ at one-sided level $.05$.

Local shift: choose $\kappa h = z_{.95} \approx 1.645$. The alternative mean equals the cutoff, so power is $.5$.

A 5% two-sided test instead uses $|Z| > z_{.975} \approx 1.96$.

Two-sided local power

For each fixed h, suppose under $P_{n,h}$ that

$$T_n \rightsquigarrow N(\kappa h, 1), \qquad c_n^{(2)} \xrightarrow{\mathbb{P}} a = z_{1-\alpha/2}.$$

Rejection rule: reject when $|T_n| > c_n^{(2)}$.

Limiting power

$$\begin{aligned}\beta_{\text{two}}(h) &= \lim_{n\to\infty} P_{n,h}(|T_n| > c_n^{(2)}) \\ &= 1 - \Phi(a - \kappa h) + \Phi(-a - \kappa h).\end{aligned}$$

At the null, $\beta_{\text{two}}(0) = \alpha$. The power is the same at h and $-h$.

Source: Proof: Notes, Theorem 1.7; same argument as the one-sided case

Studentized estimator tests

Corollary 4

For each fixed h, suppose under $P_{n,h}$ that

$$\sqrt{n}(\widehat{\theta}_n - \theta_n(h)) \rightsquigarrow N(0, V), \qquad \widehat{V}_n \xrightarrow{\mathbb{P}} V > 0.$$

Then

$$T_n = \frac{\sqrt{n}(\widehat{\theta}_n - \theta_0)}{\sqrt{\widehat{V}_n}} \rightsquigarrow N(h/\sqrt{V}, 1).$$

Assign any fixed value when $\widehat{V}_n \leq 0$.

- **Local slope:** $\kappa = 1/\sqrt{V}$.
- **Power:** Theorem 3 gives $\beta(h) = 1 - \Phi(z_{1-\alpha} - h/\sqrt{V})$.

Proof of studentization

Proof of Corollary 4.

Work under $P_{n,h}$. Put $Z_n = \sqrt{n}(\widehat{\theta}_n - \theta_n(h))$.

- **Center:** $Z_n = O_{\mathbb{P}}(1)$ by Lemma 22, part 1.
- **Control the reciprocal:** define the continuous function

$$g(v) = (\max\{v, V/2\})^{-1/2}.$$

  Lemma 24 gives $g(\widehat{V}_n) = V^{-1/2} + o_{\mathbb{P}}(1)$.
- **Combine:** $g(\widehat{V}_n) = \widehat{V}_n^{-1/2}$ when $\widehat{V}_n \geq V/2$, while $P_{n,h}(\widehat{V}_n < V/2) \to 0$. Thus

$$T_n = (Z_n + h)g(\widehat{V}_n) + o_{\mathbb{P}}(1) \rightsquigarrow N(h/\sqrt{V}, 1).$$

  Use Lemma 22, parts 3 and 4.

□

A location model with finite variance

Let $X_i = \theta + \varepsilon_i$, with iid mean-zero errors of variance $\sigma^2 \in (0, \infty)$, and set $\theta_0 = 0$.

Proposition 5

For $\widehat{\sigma}_n^2 = n^{-1} \sum_i (X_i - \overline{X}_n)^2$, *under* $P_{n,h}$,

$$T_n^{\text{mean}} = \sqrt{n}\overline{X}_n / \widehat{\sigma}_n \rightsquigarrow N(h/\sigma, 1).$$

Proof of Proposition 5.

- **Translate:** $\sqrt{n}\overline{X}_n = h + n^{-1/2} \sum_i \varepsilon_i$.
- **Estimate scale:** $\widehat{\sigma}_n^2 = P_n \varepsilon^2 - \overline{\varepsilon}_n^2 \xrightarrow{\mathbb{P}} \sigma^2$.
- **Studentize:** use Lemma 25, parts 2 and 1, and Corollary 4.

□

Sample size for a specified power

For a small positive shift δ, use the mean-test approximation.

- **Target power:** solve

$$1 - \Phi(z_{1-\alpha} - \sqrt{n}\delta/\sigma) = 1 - \eta.$$

- **Required sample size:**

$$\boxed{n \approx \frac{\sigma^2 (z_{1-\alpha} + z_{1-\eta})^2}{\delta^2}.}$$

- **Example:** $\alpha = .05$, target $.80$, $\sigma = 1$, $\delta = .25$ give $n \approx 98.92$, rounded up to 99.

The sign test in a location model

Let $X_i = \theta + \varepsilon_i$. The error cdf F is continuous, symmetric, and differentiable at zero, with $f(0) > 0$.

- **Statistic:** count positive observations:

$$K_n = \sum_i \mathbf{1}\{X_i > 0\}, \qquad T_n^{\text{sign}} = 2(K_n - n/2)/\sqrt{n}.$$

- **Null law:** at $\theta = 0$, $K_n \sim \text{Bin}(n, 1/2)$.

- **Local shift:** under $\theta_n = h/\sqrt{n}$,

$$p_n = F(h/\sqrt{n}) = \tfrac{1}{2} + f(0)h/\sqrt{n} + o(n^{-1/2}).$$

No moment assumption is needed for this test.

Source: van der Vaart (1998), §14.1; location example

Local power of the sign test

Proposition 6

Under the preceding assumptions, for each fixed h under $P_{n,h}$,

$$T_n^{\text{sign}} \rightsquigarrow N(2f(0)h, 1).$$

Its one-sided local power is

$$\beta_{\text{sign}}(h) = 1 - \Phi(z_{1-\alpha} - 2f(0)h).$$

- **Local slope:** $\kappa_{\text{sign}} = 2f(0)$.
- **Interpretation:** $f(0)$ controls how much probability crosses zero under a small shift.

Proof of the sign-test limit

Proof of Proposition 6.

Work under $P_{n,h}$, where $K_n \sim \mathrm{Bin}(n, p_n)$ and $p_n \to 1/2$.

- **Row limit:** Lemma 23 gives

$$Z_n = \frac{K_n - np_n}{\sqrt{np_n(1-p_n)}} \rightsquigarrow N(0,1).$$

- **Decompose:**

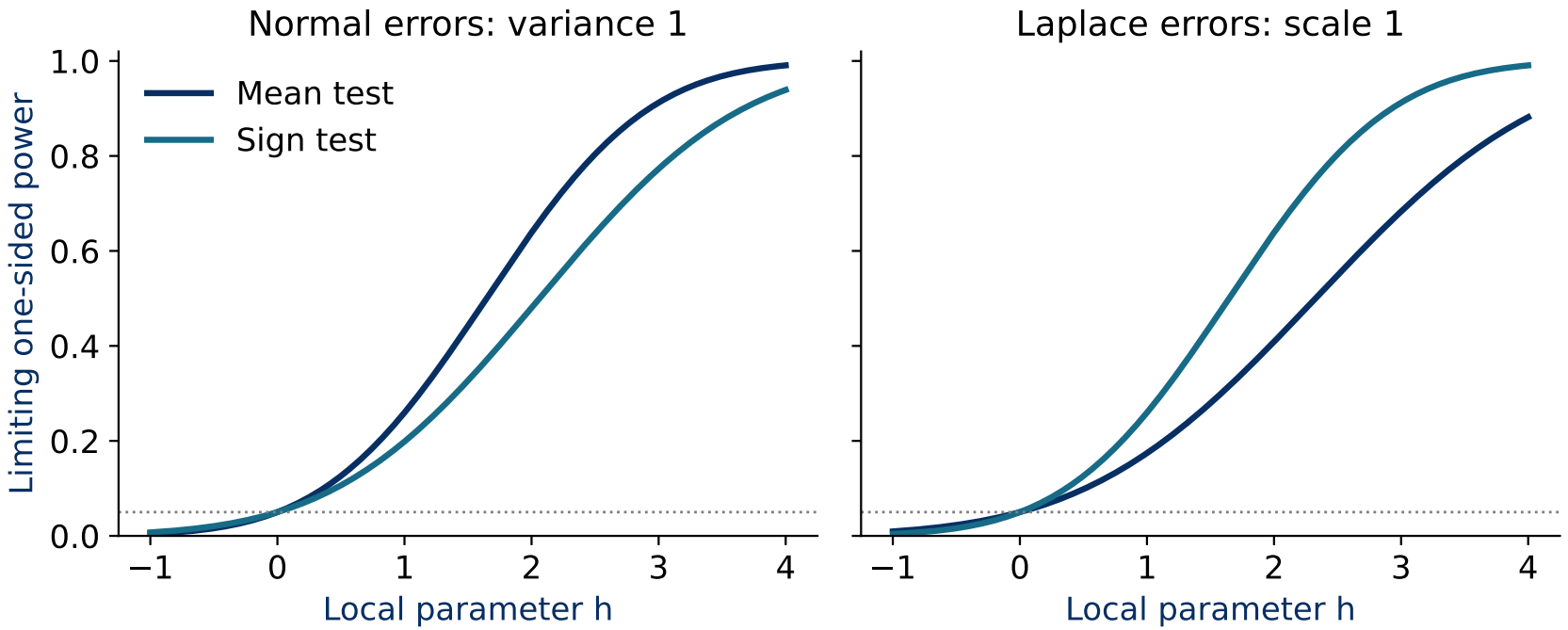
$$T_n^{\mathrm{sign}} = 2\sqrt{p_n(1-p_n)}\,Z_n + 2\sqrt{n}(p_n - 1/2).$$

  The multiplier tends to 1; the shift tends to $2f(0)h$.

- **Power:** use Lemma 22, part 4, continuous mapping in part 2, and Theorem 3.

□

Mean and sign local power



Both panels use $\alpha = .05$. The local slopes change with the error distribution. All curves are asymptotic illustrations.

The orientation of ARE

Fix a common level α and target limiting power $1-\eta \in (\alpha, 1)$. Here η is the target limiting Type II error probability.

Definition 7 (Pitman ARE)

For $j = A, B$, let $n_j(\delta)$ be sample sizes calibrated to limiting power $1-\eta$ at $\theta_0 + \delta$ as $\delta \downarrow 0$.

$$\mathrm{ARE}(A, B) = \lim_{\delta \downarrow 0} \frac{n_B(\delta)}{n_A(\delta)},$$

when the limit exists.

- **Above one:** A needs fewer observations.
- **Below one:** A needs the reciprocal sample-size multiplier.

Source: van der Vaart (1998), §14.3; sample-size convention stated explicitly

The squared-slope formula

Theorem 8 (Pitman sample-size comparison)

Fix a common level α and target limiting power $1-\eta$. For $j = A, B$, let $\kappa_j > 0$ and suppose

1. $\sqrt{n_j(\delta)}\delta \to h_j \in (0, \infty)$;
2. *along these sequences, power tends to the common target:*

$$1 - \Phi(z_{1-\alpha} - \kappa_j h_j) = 1 - \eta \in (\alpha, 1).$$

Then $\mathrm{ARE}(A, B) = \kappa_A^2 / \kappa_B^2$.

In plain language. Twice the slope needs $1/4$ as many observations.

Source: van der Vaart (1998), §14.3; sequential formulation

Proof of the squared-slope formula

Proof of Theorem 8.

- **Match power:** quantile inversion gives

$$\kappa_j h_j = z_{1-\alpha} + z_{1-\eta} =: C > 0.$$

  Here $C > 0$ because $1 - \eta > \alpha$.

- **Recover sample size:**

$$n_j(\delta)\delta^2 \to C^2/\kappa_j^2.$$

- **Take the ratio:**

$$\frac{n_B(\delta)}{n_A(\delta)} \to \frac{C^2/\kappa_B^2}{C^2/\kappa_A^2} = \frac{\kappa_A^2}{\kappa_B^2}.$$

□

Sign-test ARE relative to the mean

In $X_i = \theta + \varepsilon_i$, both tests concern θ. Errors have symmetric density f and variance $\sigma^2 < \infty$.

- **Slopes:** $\kappa_{\text{mean}} = 1/\sigma$, $\kappa_{\text{sign}} = 2f(0)$.
- **Sample-size comparison:** Theorem 8 gives

$$\boxed{\text{ARE}(\text{sign}, \text{mean}) = 4\sigma^2 f(0)^2.}$$

- **Distribution matters:** sign wins if $2\sigma f(0) > 1$. At fixed variance, a higher $f(0)$ favors sign.

Quick check. If the ARE is 0.64, what is the approximate sample-size multiplier for the sign test?

Normal errors

For $\varepsilon \sim N(0, \sigma^2)$, $f(0) = 1/(\sigma\sqrt{2\pi})$.

- **Efficiency:** $\mathrm{ARE}(\mathrm{sign}, \mathrm{mean}) = 2/\pi \approx .637$.

- **Sample-size multiplier:** $n_{\mathrm{sign}}/n_{\mathrm{mean}} \approx \pi/2 \approx 1.571$.

At level $.05$, target power $.80$, $\sigma = 1$, and shift $.25$:

Test	Planned n	Calculation
Mean	99	Exact normal, known variance
Sign	156	Local power approximation

Why mean wins: signs discard informative magnitudes.

Laplace and logistic errors

Here f is the error density and F its cdf ($x \in \mathbb{R}$).

Error law	Variance σ^2	$f(0)$	Sign/mean ARE
Laplace ($b > 0$) $f(x) = (2b)^{-1} e^{-\lvert x \rvert / b}$	$2b^2$	$1/(2b)$	2
Logistic ($s > 0$) $F(x) = (1 + e^{-x/s})^{-1}$	$\pi^2 s^2/3$	$1/(4s)$	$\pi^2/12 \approx .822$

- **Laplace:** sign needs half as many observations as mean.
- **Logistic:** sign needs $12/\pi^2 \approx 1.22$ times as many observations as mean.

Finite-sample calculation: normal location

Use normal errors, $n = 100$, $\theta = .25$, $\sigma = 1$, and level $.05$.

Test and calculation	Power
Mean, exact normal	$.804$
Sign, exact randomized binomial	$.632$
Sign, local normal approximation	$.637$

- **Exact sign rule:** reject above 58 successes; at 58, reject with probability $\rho = .255109$.
- **Calibration:** the derivation is in Notes, Proposition A.1.

Source: Direct calculations for this lecture; rounded values

An exercise in local power and sample size

For $\theta_n = h/\sqrt{n}$, suppose $T_n \rightsquigarrow N(1.5h, 1)$ under $P_{n,h}$.

Questions

1. Find local power at level $.05$ and $h = 1$.
2. Compare with a test of local slope 1. State the ARE orientation.
3. Find the planned n for shift $.2$ and target power $.80$.

Quick check. Work for two minutes. Write the sample-size ratio before calculating the ARE.

What remains to be proved

We have converted a local slope into a sample-size comparison.

- **Local distributions:** how can the likelihood supply them beyond location examples?
- **Power bound:** what is the largest possible local slope?
- **Procedures:** when do score, Wald, and LR tests attain that information limit?

Next lecture: local asymptotic normality and a Gaussian experiment.

Source: van der Vaart (1998), Chapter 6, §§7.1–7.3, and §§15.1–15.3

Second lecture

LAN, optimality, and likelihood tests

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Local likelihood and optimal testing

Goal: find the largest local power and the tests that attain it.

1. **Likelihood:** obtain a local Gaussian approximation.
2. **Change of measure:** derive limits under local alternatives.
3. **Representation:** compare all sample tests with Gaussian tests.
4. **Optimality:** apply the Gaussian Neyman–Pearson bound.
5. **Likelihood tests:** connect the score, fitted displacement, and gain in fit.

Source: van der Vaart (1998), Chapters 6–7 and §§15.1–15.3

Likelihood, score, and information

For density p_θ relative to μ, define

- **One observation:**

$$\ell_\theta = \log p_\theta, \quad s_\theta = \nabla \ell_\theta, \quad I(\theta) = \mathbb{E}_\theta[s_\theta s_\theta^T].$$

- **Full sample:** $\ell_n(\theta) = nP_n\ell_\theta$ and total information is $nI(\theta)$.

- **Central sequence:** write $s_0 = s_{\theta_0}$ and $I = I(\theta_0)$:

$$\Delta_n = n^{-1/2} \sum_i s_0(X_i).$$

For a centered score with finite second moments, the Lemma 25, part 2 gives $\Delta_n \rightsquigarrow N_d(0, I)$ under $P_{n,0}$.

Source: van der Vaart (1998), Chapter 7; Notes Section 3.1

The change-of-measure identity

Assume nearby distributions have a common positive support.

- **Likelihood ratio:**

$$L_{n,h} = \frac{dP_{n,h}}{dP_{n,0}} = e^{\ell_n(\theta_n(h)) - \ell_n(\theta_0)}.$$

- **Expectation under an alternative:** for bounded measurable g,

$$\boxed{\mathbb{E}_{n,h} g(T_n) = \mathbb{E}_{n,0}[g(T_n) L_{n,h}].}$$

  This is the defining integral identity for a density of one measure relative to another; see Lemma 26, part 2.

To pass to a limit, we must control the likelihood ratio's tails.

Source: van der Vaart (1998), Chapter 6; Notes Section 3.1

A statistical experiment

A statistical experiment is a measurable sample space with an indexed family of probability laws.

- **Local experiment:**

$$\mathcal{E}_n = \{P_{n,h} : h \in H_n\}, \quad H_n = \{h : \theta_0 + h/\sqrt{n} \in \Theta\}.$$

  Every fixed h belongs to H_n eventually.

- **Candidate Gaussian experiment:** for $I > 0$,

$$\mathcal{G} = \{Q_h = N_d(h, I^{-1}) : h \in \mathbb{R}^d\}.$$

  We observe one Gaussian vector with unknown mean h.

Source: van der Vaart (1998), §§7.1–7.3

The exact Gaussian likelihood ratio

Let $Y \sim Q_h = N_d(h, I^{-1})$, with $I > 0$.

- **Subtract the two log densities:**

$$\log \frac{dQ_h}{dQ_0}(Y) = -\tfrac{1}{2}(Y-h)^T I(Y-h) + \tfrac{1}{2}Y^T IY$$
$$= h^T IY - \tfrac{1}{2}h^T Ih.$$

- **Under the null:** $IY \sim N_d(0, I)$.

- **Form to recognize:** a random linear term and a deterministic quadratic term.

Local asymptotic normality

Definition 9 (LAN at θ_0)

For each fixed h, under $P_{n,0}$,

$$\log L_{n,h} = h^T \Delta_n - \tfrac{1}{2} h^T I h + o_{\mathbb{P}}(1), \qquad \Delta_n \rightsquigarrow N_d(0, I),$$

where $I > 0$. Nearby local laws are absolutely continuous with respect to $P_{n,0}$, so these are genuine likelihood ratios.

- **Gaussian form:** the same quadratic likelihood as the preceding experiment.
- **Local observation:** $I^{-1}\Delta_n$ will have limit $N_d(h, I^{-1})$ under $P_{n,h}$; see Theorem 13.

Source: van der Vaart (1998), Chapter 7

Smoothness assumptions for LAN

Take an open ball $\mathcal{U}$ around the interior point θ_0.

1. **Support and derivatives:** p_θ is positive on a common support; $\ell_\theta(x)$ has three continuous derivatives on $\mathcal{U}$.
2. **Moments and curvature:** with $H_0 = \nabla^2 \ell_{\theta_0}$,

$$\mathbb{E}_0 s_0 = 0, \quad I = \mathbb{E}_0 s_0 s_0^T = -\mathbb{E}_0 H_0 > 0.$$

   Also $\mathbb{E}_0 \|s_0\|^2 + \mathbb{E}_0 \|H_0\| < \infty$.

3. **Remainder control:** $\sup_{\theta \in \mathcal{U}} \|\nabla^3 \ell_\theta(X)\| \le M(X)$ with $\mathbb{E}_0 M < \infty$.

Theorem 10 (Smooth iid LAN)

These conditions give LAN with score central sequence and a remainder uniform on each bounded set of h.

Source: Sufficient-condition version for this lecture; van der Vaart, Chapter 7

Proof of smooth iid LAN

Proof of Theorem 10.

Fix $K < \infty$ and work under $P_{n,0}$.

- **Taylor expansion:** writing $\overline{H}_n = P_n H_0$,

$$\ell_n(\theta_0 + h/\sqrt{n}) - \ell_n(\theta_0) = h^T \Delta_n + \tfrac{1}{2} h^T \overline{H}_n h + R_n(h).$$

- **Uniform error:** for $\|h\| \le K$, it is bounded by

$$\tfrac{1}{2} K^2 \|\overline{H}_n + I\| + \frac{K^3}{6\sqrt{n}} P_n M \xrightarrow{\mathbb{P}} 0.$$

  Use the Lemma 25, part 1.

- **Central sequence:** the Lemma 25, part 2 gives $\Delta_n \rightsquigarrow N_d(0, I)$.

□

Worked example: Bernoulli likelihood

Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, $p_0 \in (0,1)$.

- **Score and information:**

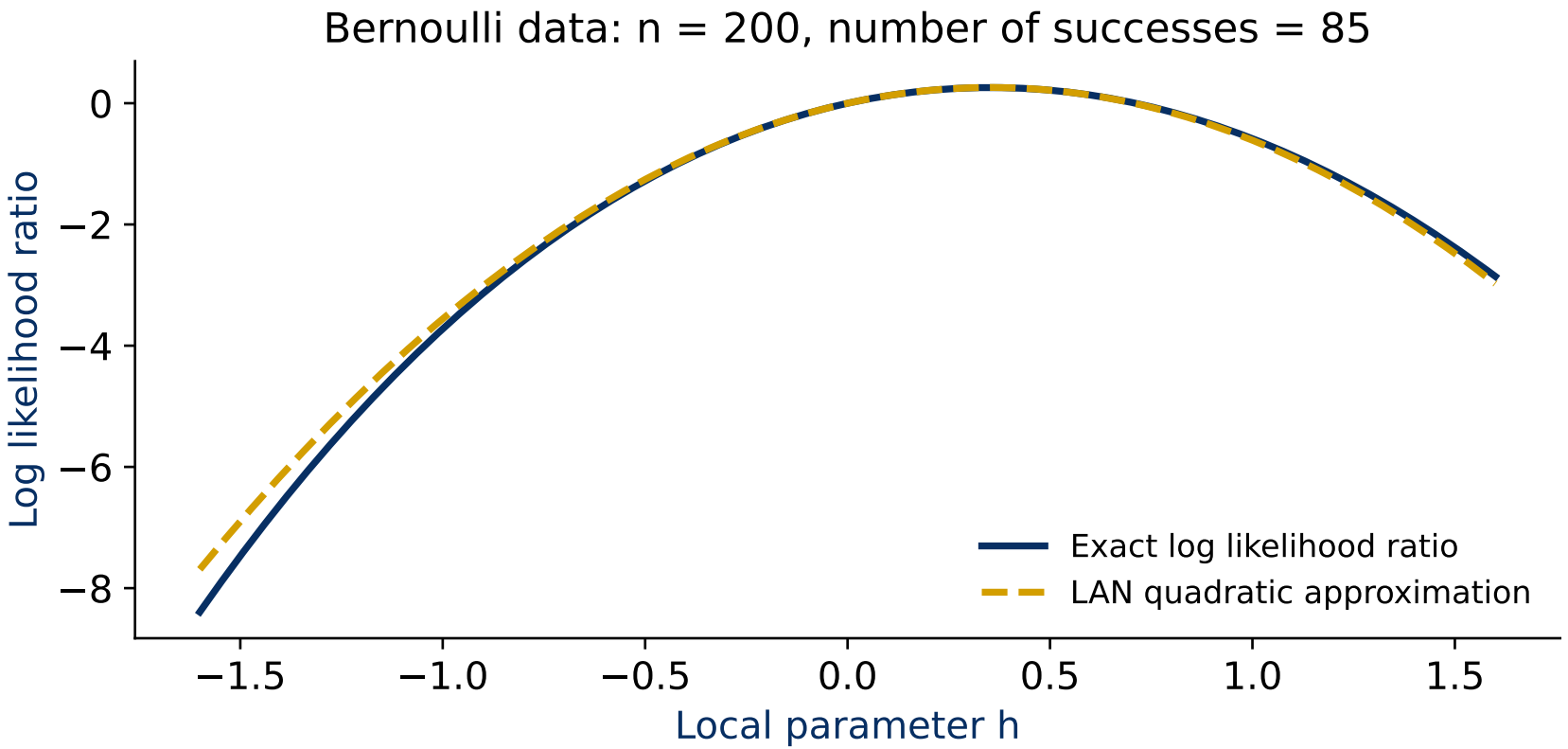
$$s_0(x) = \frac{x - p_0}{p_0(1-p_0)}, \qquad I = \frac{1}{p_0(1-p_0)}.$$

- **Local likelihood:** with $\Delta_n = \sqrt{n}(\overline{X}_n - p_0)/[p_0(1-p_0)]$,

$$\log L_{n,h} = h\Delta_n - \frac{h^2}{2p_0(1-p_0)} + o_{\mathbb{P}}(1).$$

- **Conditions:** the support is $\{0,1\}$; derivatives are bounded on a small closed interval around p_0. Apply Theorem 10.

The local quadratic approximation



The approximation concerns the likelihood as a function of the local parameter. The original observations remain Bernoulli.

Contiguity

Definition 11

$\mathsf{Q}_n \triangleleft \mathsf{P}_n$ means that, for every sequence of measurable events,

$$\mathsf{P}_n(A_n) \to 0 \quad \Longrightarrow \quad \mathsf{Q}_n(A_n) \to 0.$$

- **LAN conclusion:** $P_{n,h} \triangleleft P_{n,0}$ for every fixed h; see Theorem 13.
- **Remainders:** apply the definition to $A_n = \{\|r_n\| > \epsilon\}$ to transfer $r_n = o_{\mathbb{P}}(1)$.
- **Distributions:** likelihood tilting determines how the limiting mean changes.

Source: van der Vaart (1998), Chapter 6

Likelihood tilting and contiguity

Lemma 12 (Likelihood tilting)

Let $L_n = d\mathsf{Q}_n/d\mathsf{P}_n$. *Suppose under* P_n,

$$(T_n, L_n) \rightsquigarrow (T, L), \qquad L \geq 0, \qquad \mathbb{E}L = 1.$$

Then, for bounded continuous g,

$$\mathbb{E}_{\mathsf{Q}_n} g(T_n) \to \mathbb{E}[g(T)L], \qquad \mathsf{Q}_n \triangleleft \mathsf{P}_n.$$

- **Normalization:** $\mathbb{E}L = 1$ makes the limit a probability law.
- **Tail control:** likelihood mass cannot escape to rare events.

Source: van der Vaart (1998), Chapter 6; a form of Le Cam's likelihood lemmas

Proof of Lemma 12

Proof of Lemma 12.

- **Tail bound:** $\mathbb{E}L_n = \mathbb{E}L = 1$ gives $\mathbb{E}_{\mathsf{P}_n}(L_n - K)_+ \to \mathbb{E}(L-K)_+$.

$$\limsup_n \mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > 2K\}] \le 2\mathbb{E}(L-K)_+ \to 0.$$

  Finite early indices also have vanishing tails.

- **Truncate:** weak convergence applies to $g(T_n)(L_n \wedge K)$. Remove K using the tail bound and change of measure.

- **Events:** if $\mathsf{P}_n(A_n) \to 0$, use

$$\mathsf{Q}_n(A_n) \le K\mathsf{P}_n(A_n) + \mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > K\}].$$

  Let $n \to \infty$, then $K \to \infty$.

□

The central sequence under local shifts

Theorem 13 (Gaussian shift under LAN)

For every fixed h, $P_{n,h} \triangleleft P_{n,0}$ and, under $P_{n,h}$,

$$\Delta_n \rightsquigarrow N_d(Ih, I), \qquad I^{-1}\Delta_n \rightsquigarrow N_d(h, I^{-1}).$$

- **Shifted mean:** information turns h into the score mean Ih.
- **Same covariance:** local alternatives preserve the first-order noise covariance I.

Proof of the Gaussian shift

Proof of Theorem 13.

- **Joint null limit:** LAN and Lemma 22 give

$$(\Delta_n, L_{n,h}) \rightsquigarrow (Z, e^{h^T Z - h^T I h/2}), \quad Z \sim N_d(0, I).$$

- **Normalized tilt:** the Gaussian moment-generating function makes the exponential's mean one. Apply Lemma 12.

- **Complete the square:** the tilted exponent is

$$-\tfrac{1}{2} z^T I^{-1} z + h^T z - \tfrac{1}{2} h^T I h = -\tfrac{1}{2}(z - Ih)^T I^{-1}(z - Ih).$$

This is $N_d(Ih, I)$. Multiplication by I^{-1} gives the second limit.

□

Local normality of the MLE

Corollary 14

Assume $\sqrt{n}(\widehat{\theta}_n - \theta_0) = I^{-1}\Delta_n + r_n$ *under* $P_{n,0}$, *with* $r_n = o_{\mathbb{P}}(1)$. *Under* $P_{n,h}$,

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \rightsquigarrow N_d(h, I^{-1}), \qquad \sqrt{n}(\widehat{\theta}_n - \theta_n(h)) \rightsquigarrow N_d(0, I^{-1}).$$

Proof of Corollary 14.

- **Transfer the error:** contiguity in Theorem 13 gives $r_n = o_{\mathbb{P}}(1)$ under $P_{n,h}$.
- **Shift the limit:** use that theorem and Lemma 22, part 4, then subtract h.

□

Asymptotic representation of tests

Theorem 15 (Representation in a LAN experiment)

For any tests $0 \le \phi_n \le 1$, every subsequence has a further subsequence and a measurable test $\phi_\infty : \mathbb{R}^d \to [0,1]$ such that

$$\mathbb{E}_{n,h}\phi_n \to \mathbb{E}_{Q_h}\phi_\infty(Y), \qquad Q_h = N_d(h, I^{-1}),$$

for every fixed h along that further subsequence.

- **One representation:** the same subsequence and test work for all fixed h.
- **Scope:** this represents power functions. It does not assert uniform convergence over h.

Source: van der Vaart (1998), §15.1; LAN-specialized formulation

Proof of the representation theorem

Proof of Theorem 15.

- **Extract under the null:** tightness and Lemma 26, part 1 give

$$(\Delta_n, \phi_n) \rightsquigarrow (Z, U), \quad Z \sim N_d(0, I), \quad 0 \le U \le 1.$$

- **Average remaining randomness:** Lemma 26, part 2 constructs $r(z) = \mathbb{E}[U \mid Z = z]$. Set $\phi_\infty(y) = r(Iy)$.

- **Tilt:** for each fixed h, LAN and Lemma 12 give

$$\mathbb{E}_{n,h}\phi_n \to \mathbb{E}[U e^{h^T Z - h^T I h/2}] = \mathbb{E}_{Q_h}\phi_\infty(Y).$$

  Use the bounded continuous clipped function $g(u) = \max(0, \min(1, u))$.

□

Gaussian one-sided optimality

Proposition 16 (Gaussian UMP test)

For $Y \sim N(h, I^{-1})$, the uniformly most powerful (UMP) level-α test of $h \le 0$ against $h > 0$ is

$$\phi^*_\infty(Y) = \mathbf{1}\{\sqrt{I}\, Y > z_{1-\alpha}\}.$$

Its power is $\beta^*(h) = 1 - \Phi(z_{1-\alpha} - h\sqrt{I})$.

- **Level:** power increases in h and equals α at zero.
- **Optimality:** the likelihood ratio increases in Y for each $h > 0$; see Notes, Proposition 4.4.

Source: van der Vaart (1998), §15.2; proof in Notes, Proposition 4.4

A local asymptotic power bound

Theorem 17 (One-sided power envelope)

In a scalar LAN model with $I > 0$, any tests satisfying $\limsup_n \mathbb{E}_{n,0}\phi_n \le \alpha$ obey, for every fixed $h > 0$,

$$\limsup_n \mathbb{E}_{n,h}\phi_n \le 1 - \Phi(z_{1-\alpha} - h\sqrt{I}).$$

Proof of Theorem 17.

1. Choose a subsequence attaining the limsup at this h.
2. Theorem 15 gives a Gaussian test of size at most α at zero.
3. Proposition 16 bounds its power by $\beta^*(h)$.

□

Source: van der Vaart (1998), §15.3

A locally AUMP score test

Local asymptotic level: $\limsup_n \mathbb{E}_{n,h}\phi_n \le \alpha$ for every fixed $h \le 0$.

Local AUMP: asymptotically uniformly most powerful means attaining a limiting power that dominates every such competitor at each fixed $h > 0$.

Theorem 18

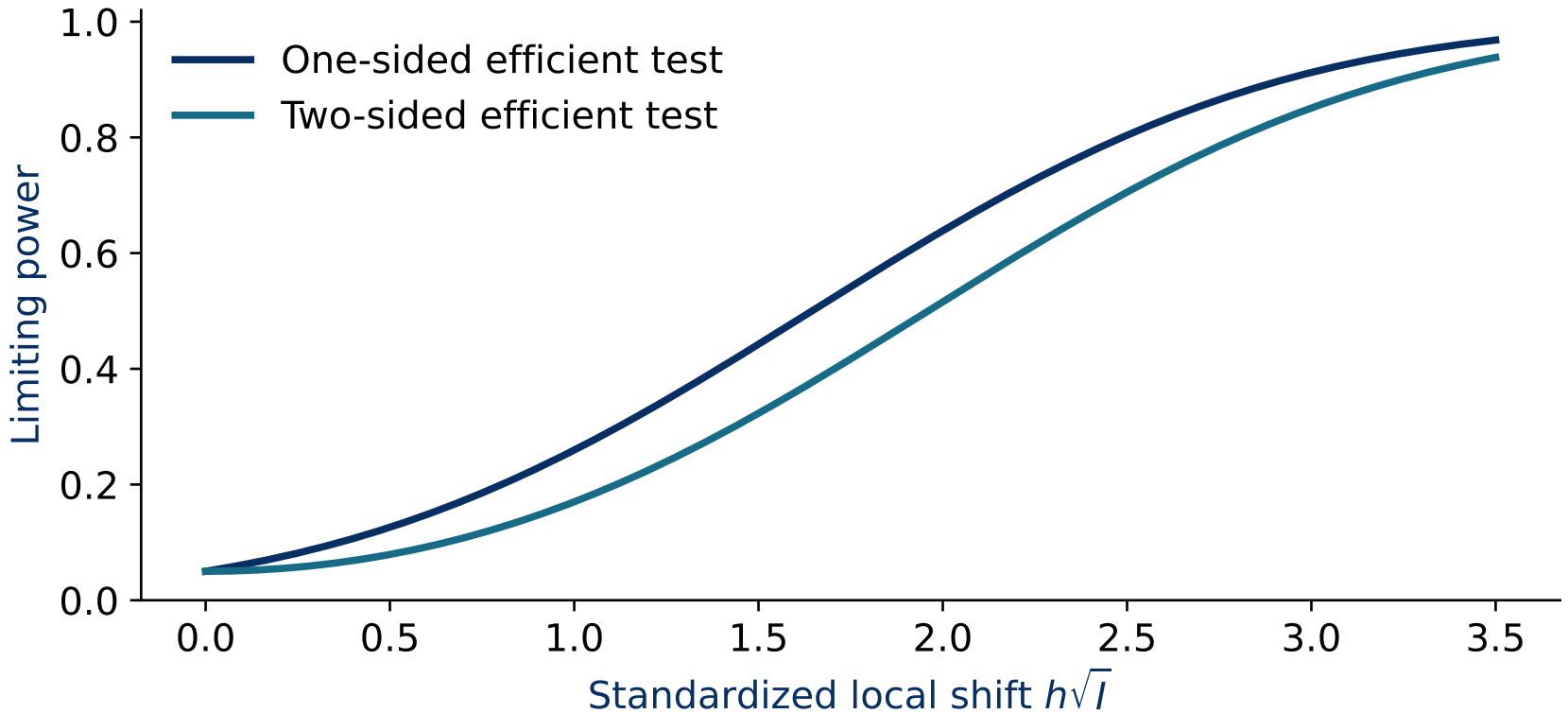
The score test $\mathbf{1}\{\Delta_n/\sqrt{I} > z_{1-\alpha}\}$ is locally AUMP.

Proof of Theorem 18.

- **Power:** Theorems 13 and 3 give $\beta^*(h)$.
- **Level and bound:** $\beta^*(h) \le \alpha$ for $h \le 0$; for $h > 0$ it attains Theorem 17.

□

One-sided and two-sided power



Write $\chi^2_1(\lambda)$ for the law of $(Z + \sqrt{\lambda})^2$, $Z \sim N(0, 1)$. Efficient two-sided tests have $\lambda = h^2 I$. The one-sided AUMP result concerns the positive direction.

Rao's score statistic

For scalar $H_0 : \theta = \theta_0$, let $U_n(\theta_0) = \sum_i s_0(X_i)$.

- **Signed statistic:** use the likelihood slope at the null:

$$Z_{S,n} = \frac{U_n(\theta_0)}{\sqrt{nI}} = \frac{\Delta_n}{\sqrt{I}}.$$

- **Local law:** Theorem 13 gives $Z_{S,n} \rightsquigarrow N(h\sqrt{I}, 1)$ under $P_{n,h}$.

- **Two-sided version:** by Lemma 22, part 2,

$$S_n = Z_{S,n}^2 \rightsquigarrow \chi_1^2(h^2 I).$$

Source: Lehmann and Romano (2005), 3rd ed., §12.4.3

Wald's statistic

Assume the efficient expansion and $\widehat{I}_n \xrightarrow{\mathbb{P}} I > 0$ under $P_{n,0}$.

- **Signed statistic:** standardize the fitted displacement:

$$Z_{W,n} = \sqrt{n\widehat{I}_n}(\widehat{\theta}_n - \theta_0).$$

- **Local law:** Corollary 14 transfers the expansion and information consistency. Under $P_{n,h}$,

$$Z_{W,n} \rightsquigarrow N(h\sqrt{I}, 1).$$

- **Two-sided version:** $W_n = Z_{W,n}^2 \rightsquigarrow \chi_1^2(h^2 I)$ by Lemma 22, part 2.

Source: Lehmann and Romano (2005), 3rd ed., §12.4.2

Score and Wald equivalence

Proposition 19

Under the efficient expansion and consistent information, for every fixed h under $P_{n,h}$,

$$Z_{W,n} - Z_{S,n} = o_{\mathbb{P}}(1), \qquad W_n - S_n = o_{\mathbb{P}}(1).$$

Proof of Proposition 19.

- **Signed error:** Corollary 14 and Lemma 22, part 3 give

$$Z_{W,n} - Z_{S,n} = (\sqrt{\widehat{I}_n}/I - I^{-1/2})\Delta_n + \sqrt{\widehat{I}_n} r_n = o_{\mathbb{P}}(1).$$

- **Squares:** multiply this error by the tight sum $Z_{W,n} + Z_{S,n}$. Contiguity in Theorem 13 transfers both conclusions.

□

The efficient estimator expansion

Lemma 20 (Consistent interior score root)

Under Theorem 10, assume $\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ *and an interior score root* $U_n(\widehat{\theta}_n) = 0$ *with probability tending to one. Then*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = I^{-1}\Delta_n + o_{\mathbb{P}}(1).$$

- **Rate:** root-n consistency follows from the score equation; see Notes, Lemma 5.2.

- **Influence function:** $\psi_\theta = s_\theta$, $A = -I$, $B = I$, so $\varphi = I^{-1}s_0$ and $V = I^{-1}$.

Likelihood ratio test

For a scalar point null, define

$$\Lambda_n = 2\{\ell_n(\widehat{\theta}_n) - \ell_n(\theta_0)\}.$$

Theorem 21 (Regular scalar likelihood tests)

Under Theorem 10, assume a consistent unrestricted MLE that is an interior score root with probability tending to one. Then, under $P_{n,h}$,

$$\Lambda_n = \Delta_n^2 / I + o_{\mathbb{P}}(1).$$

With consistent information, $\Lambda_n, S_n, W_n \rightsquigarrow \chi^2_1(h^2 I)$.

Common local source: slope at zero, displacement of the maximum, and twice the fitted gain.

Source: Lehmann and Romano (2005), 3rd ed., §12.4.4; see also van der Vaart, Chapter 16

Proof of the likelihood ratio expansion

Proof of Theorem 21.

- **Localize:** Lemma 20 gives $\widehat{h}_n = \sqrt{n}(\widehat{\theta}_n - \theta_0) = \Delta_n / I + o_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$.
- **Complete the square:** use uniform LAN; see Notes, Theorem 5.3:

$$\begin{aligned} \ell_n(\widehat{\theta}_n) - \ell_n(\theta_0) &= \frac{\Delta_n^2}{2I} - \frac{I}{2}(\widehat{h}_n - \Delta_n/I)^2 + o_{\mathbb{P}}(1) \\ &= \frac{\Delta_n^2}{2I} + o_{\mathbb{P}}(1). \end{aligned}$$

- **Transfer:** multiply by two. Theorem 13 transfers the error and gives the local law; Proposition 19 supplies Wald equivalence.

□

Worked example: three Bernoulli tests

Let $\widehat{p} = \overline{X}_n$ and test $p = p_0 \in (0,1)$ against $p \neq p_0$.

Test	Statistic
Wald	$n(\widehat{p} - p_0)^2 / [\widehat{p}(1-\widehat{p})]$
Score	$n(\widehat{p} - p_0)^2 / [p_0(1-p_0)]$
LR	$2n\left[\widehat{p}\log\frac{\widehat{p}}{p_0} + (1-\widehat{p})\log\frac{1-\widehat{p}}{1-p_0}\right]$

- **Local limit:** Theorem 21 gives $\chi^2_1(h^2/[p_0(1-p_0)])$.
- **Sample boundaries:** use $0\log 0 = 0$ for LR. Wald needs $0 < \widehat{p} < 1$; boundary probability vanishes locally.

Worked calculation: 52 successes in 100 trials

Test $p = .40$ against $p \neq .40$ using $n = 100$ and $\widehat{p} = .52$.

Test	Statistic	Approximate p-value
Wald	5.769	$.0163$
Score	6.000	$.0143$
LR	5.864	$.0155$

- **Calibration:** use χ^2_1, with $.95$ quantile 3.841.
- **Decision:** all reject at $.05$. First-order equivalence allows different finite-sample values.

Source: Direct likelihood calculations for this lecture

Efficiency, sample size, and information

For a scalar locally regular asymptotically linear estimator:

Quantity	Information constraint
Estimator variance	$V \geq I^{-1}$
Studentized local slope	$\kappa = 1/\sqrt{V} \leq \sqrt{I}$
ARE relative to an efficient test	$1/(VI) \leq 1$

- **Variance bound:** Notes, Proposition F.1.
- **Slope and ARE:** Corollary 4 and Theorem 8.
- **Power envelope:** Theorem 17.

Exercise: a Gaussian local experiment

Let $X_i \overset{\text{iid}}{\sim} N(\theta, 4)$ and test $\theta = 0$ against $\theta > 0$.

Questions

1. Find s_0, I, and Δ_n.
2. Write the likelihood ratio at $\theta_n = h/\sqrt{n}$.
3. Construct the locally AUMP level-$.05$ test.
4. Calculate its power at $h = 2$.

Quick check. Use the likelihood expansion to connect the statistic and the power bound.

Reading and the main conclusions

- **Likelihood:** LAN gives a Gaussian local experiment.
- **Local laws:** tilting shifts the score mean to Ih.
- **Optimality:** representation transfers the Gaussian power bound to all tests.
- **Procedures:** regular score, Wald, and LR tests share the same first-order local information.

Reading: van der Vaart (1998), Chapters 6–7 and 15; Lehmann–Romano (2005), §§12.4.2–12.4.4.

Shared limit rules

Lemma 22 (Limit calculus)

Under laws Q_n, suppose $Y_n \rightsquigarrow Y$ and $r_n \xrightarrow{\mathbb{P}} 0$.

1. ***Tightness:*** $Y_n = O_{\mathbb{P}}(1)$.
2. ***Continuous mapping:*** $g(Y_n) \rightsquigarrow g(Y)$ *for continuous* g.
3. ***Products:*** $A_n = O_{\mathbb{P}}(1)$ *and* $B_n = o_{\mathbb{P}}(1)$ *imply* $A_n B_n = o_{\mathbb{P}}(1)$.
4. ***Joint limit:*** $(Y_n, r_n) \rightsquigarrow (Y, 0)$, *hence* $Y_n + r_n \rightsquigarrow Y$ *when dimensions agree.*

The proofs are in Notes, Lemma G.1. Probability convergence under a continuous map is stated separately in Lemma 24.

Bernoulli rows with changing probabilities

Lemma 23

For each n, let $B_{n1}, \ldots, B_{nn}$ be independent Bernoulli variables with common parameter p_n. If $p_n \to p \in (0,1)$, then

$$\frac{\sum_{i=1}^{n} B_{ni} - np_n}{\sqrt{np_n(1-p_n)}} \rightsquigarrow N(0,1).$$

The statistic is defined for all sufficiently large n.

In plain language. Independence is required within each row. No independence across rows is needed.

Source: Proof: Notes, Lemma G.3

Continuous mapping in probability

Lemma 24

If $Y_n \xrightarrow{\mathbb{P}} y$ and g is continuous at y, then $g(Y_n) \xrightarrow{\mathbb{P}} g(y)$. Underlying probability spaces and laws may change with n.

The same continuity argument works under changing sample laws.

Source: Proof: Notes, Lemma G.1, part 1

Probability prerequisites

Lemma 25 (IID LLN and CLT)

Let $W_1, W_2, \ldots$ be iid random vectors in $\mathbb{R}^d$.

1. ***Law of large numbers:*** *if $\mathbb{E}\|W_1\| < \infty$,*
$$n^{-1} \sum_i W_i \xrightarrow{\mathbb{P}} \mathbb{E}W_1.$$

2. ***Central limit theorem:*** *if $\mathbb{E}\|W_1\|^2 < \infty$,*
$$\sqrt{n}(\overline{W}_n - \mathbb{E}W_1) \rightsquigarrow N_d(0, \mathrm{Cov}(W_1)).$$

Standard prerequisites, stated without proof. The sign test instead uses Lemma 23 for changing Bernoulli probabilities.

Compactness and conditional averaging

Lemma 26 (Representation tools)

1. ***Finite-dimensional Prokhorov theorem:*** *every tight sequence of probability laws on a Euclidean space has a weakly convergent subsequence.*

2. ***Radon–Nikodym theorem:*** *finite measures $m \ll \nu$ admit a measurable density r satisfying*

$$m(B) = \int_B r \, d\nu.$$

If $0 \le m(B) \le \nu(B)$ for all measurable B, choose $0 \le r \le 1$.

For the representation proof, take ν to be the law of Z and $m(B) = \mathbb{E}[U \mathbf{1}\{Z \in B\}]$. Its density is a version of $\mathbb{E}[U \mid Z = z]$.

Source: Standard measure/probability prerequisites; Notes, Section 4.1 gives the construction

Delta method and matrix inversion

Lemma 27 (Differentiability and inverse continuity)

1. ***Delta method:*** *if* $a_n \to \infty$, $a_n(\widehat{\theta}_n - \theta_0) \rightsquigarrow Z$, *and* g *is differentiable at* θ_0, *then*

$$a_n\{g(\widehat{\theta}_n) - g(\theta_0)\} \rightsquigarrow Dg(\theta_0)Z.$$

2. ***Matrix inverse:*** *if* $A_n \xrightarrow{\mathbb{P}} A$ *and* A *is nonsingular, then* A_n *is nonsingular with probability tending to one and* $A_n^{-1} \xrightarrow{\mathbb{P}} A^{-1}$, *with arbitrary values otherwise.*

If A is symmetric positive definite and A_n is symmetric, positive definiteness also holds with probability tending to one.

Source: Proofs: Notes, Lemma G.2